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Introduction to Data Communications

Topic 5 : DLL and Physical layer

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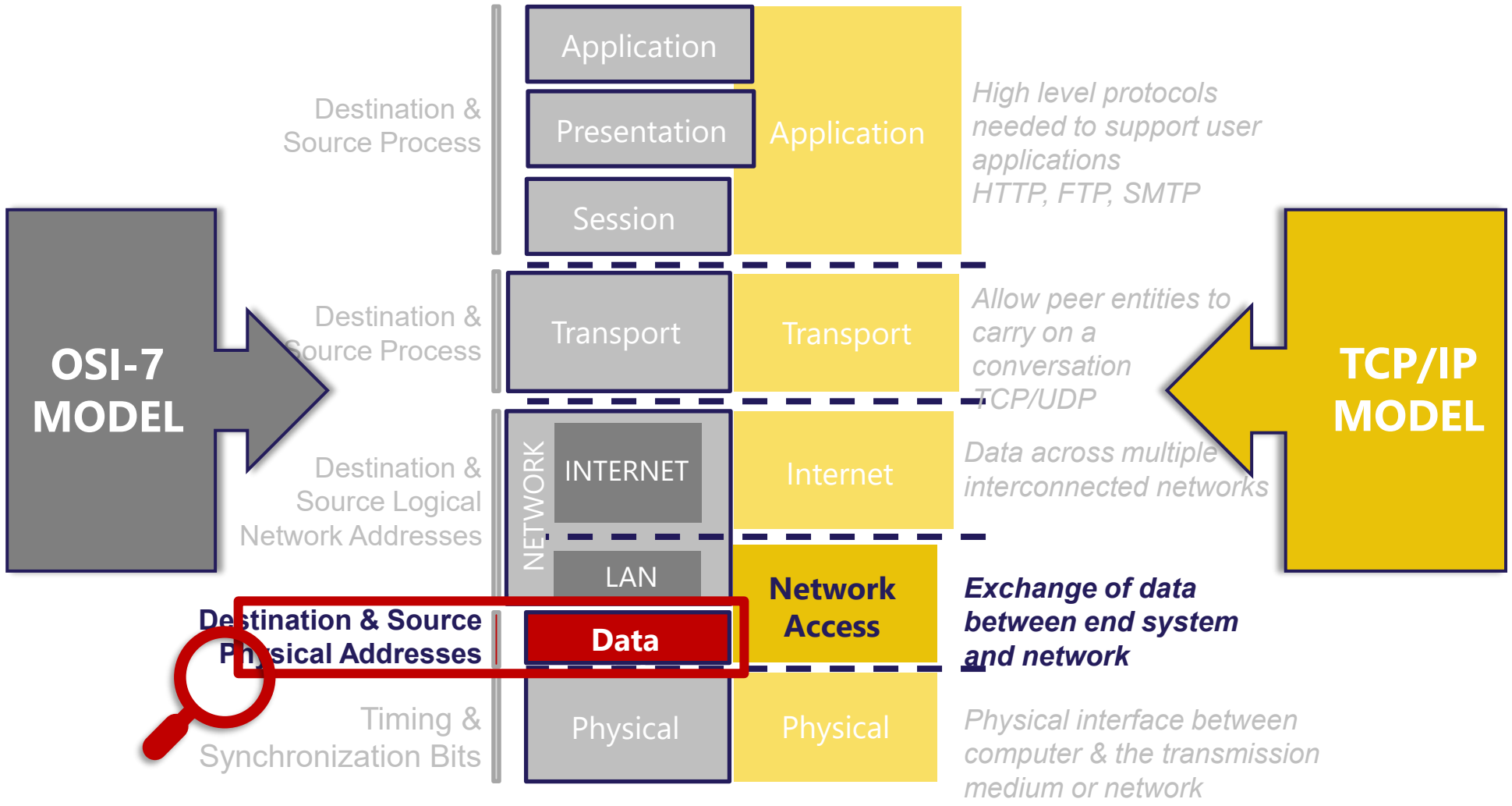
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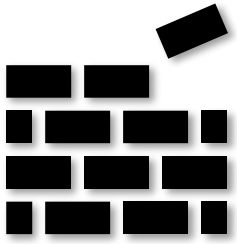


Topic 5 – Learning Outcomes

- Data-link Layer
- LLC: Flow Control
- LLC: Error Detection
- CSMA/CA and CSMA/CD
- LAN, Ethernet Fundamentals
- What is collision, broadcast domain?
- STP protocol


OSI-7 MODEL
TCP/IP MODEL
Application
Presentation
Session
Application
Destination & Source Process
High level protocols needed to support user applications
HTTP, FTP, SMTP
Transport
Transport
Destination & Source Process
Allow peer entities to carry on a conversation
TCP/UDP
NETWORK
INTERNET
Internet
Destination & Source Logical Network Addresses
Data across multiple interconnected networks
LAN
Network Access
Destination & Source Physical Addresses
Data
Exchange of data between end system and network
Physical
Physical
Timing & Synchronization Bits
Physical interface between computer & the transmission medium or network

Data Link Layer

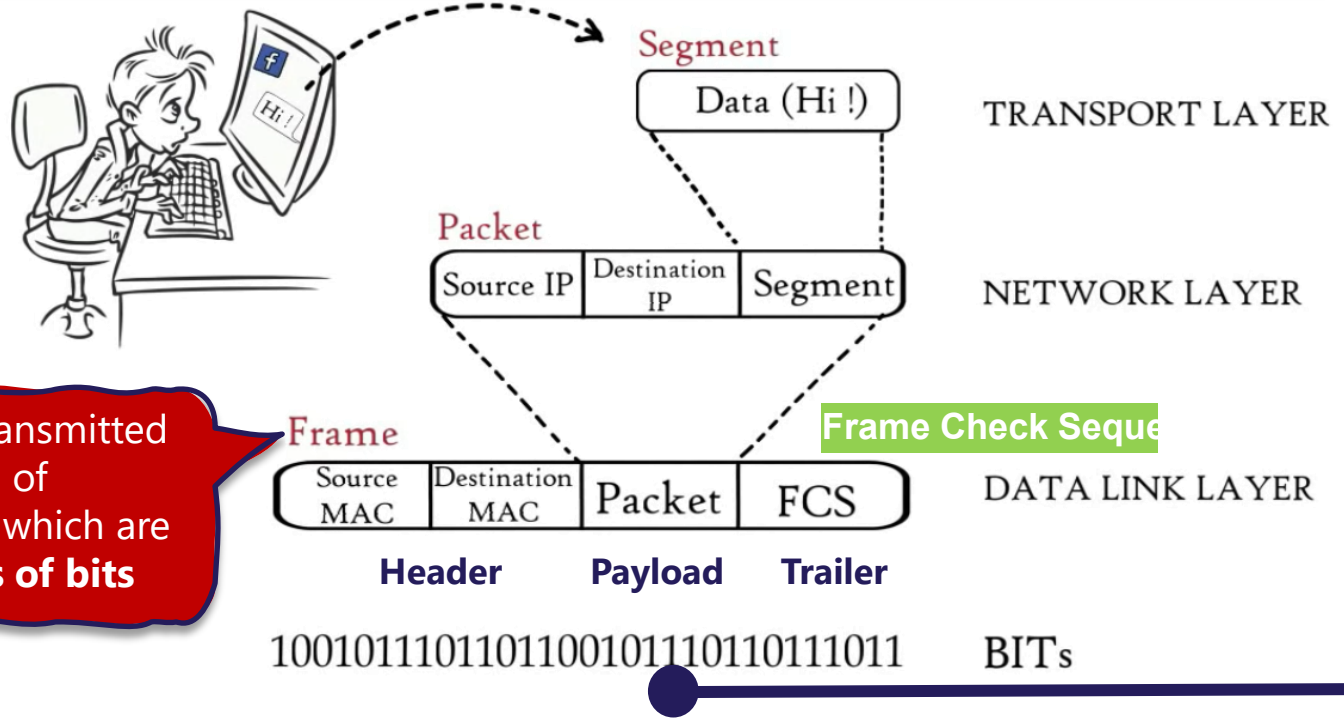


The data link layer provides the **building blocks for communication** across a variety of physical media

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Data Link Layer

The data link layer is concerned with local delivery of **frames** between nodes

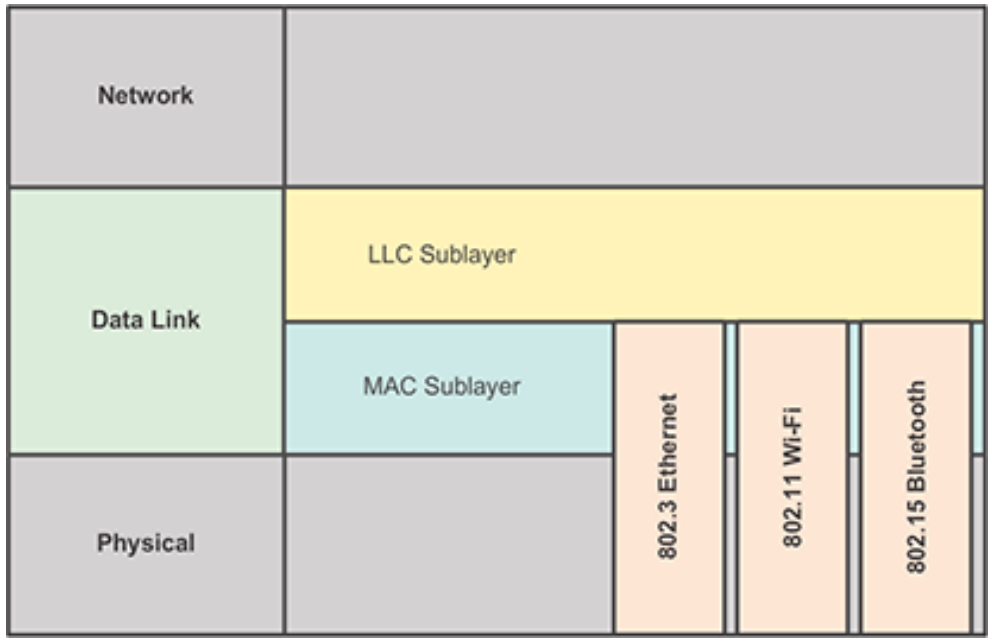


Data transmitted in units of frames which are **groups of bits**

- **Implementation**
  - Primarily implemented in software
  - May be embedded in physical devices such as switches and network adapters (firmware)

- **Data Link Layer Protocols**
  - Point-Point Protocol (PPP)
  - High-Level Data Link Control (HDLC)
  - IEEE 802.3 Ethernet LAN
  - IEEE 802.11 (Wi Fi) LAN

- **Two Sub Layers**
  1. **LLC (Logical Link Control)**
  2. **MAC (Media Access Control)**



Data Link Layer: Services

LLC:

- Provide services to network layer protocols
- **Flow Control**
- **Error Control**

MAC:

- **Framing:** bits to frame (vice versa)
- Physical addressing **(MAC addressing)**
- Multiple access methods for channel-access control **(CSMA/CD, CSMA/CA)**
- LAN switching (packet switching), including MAC filtering, **Spanning Tree Protocol (STP)**
- Data packet queuing or scheduling
- Store-and-forward switching or cut-through switching
- Quality of Service (QoS) control
- **Virtual LANs (VLAN)**

Stop-and-Wait Protocol:

- Overview: Stop-and-Wait is a simple and basic flow control protocol used in data communication. It ensures that data packets are transmitted sequentially, with each packet requiring an acknowledgment (ACK) from the receiver before the next packet can be sent.
- Operation: The sender transmits a single packet and then waits for an acknowledgment from the receiver. If the acknowledgment is not received within a specified timeout period or if there's an error, the sender retransmits the same packet.
- Advantages: Simplicity, easy to implement, suitable for scenarios with low error rates.
- Disadvantages: Inefficiency due to the stop-and-wait nature, as the sender can't utilize the network fully if the round-trip time is high

Sliding Window Protocols:

1. Overview: Sliding window protocols are a family of protocols used for reliable and sequential data transmission over a network. They allow multiple packets to be in transit simultaneously, increasing the efficiency of data transfer compared to stop-and-wait protocols.
2. Operation: The sender maintains a window of packets that it can transmit before receiving acknowledgments. The receiver sends acknowledgments for received packets, indicating which packets are accepted. Both sender and receiver maintain a window size that determines the number of packets that can be in transit at any given time.
3. Types: Sliding window protocols can be further classified into Go-Back-N ARQ and Selective Repeat ARQ.

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Go-Back-N ARQ (Automatic Repeat reQuest):

1. Overview: Go-Back-N ARQ is a sliding window protocol used for reliable data transmission in networks with high error rates, such as wireless networks.
2. Operation: In Go-Back-N ARQ, the sender can have multiple unacknowledged packets in transit. If the receiver detects an error in a packet, it discards the packet and all subsequent packets until the error is resolved. The sender receives a timeout or a cumulative acknowledgment and retransmits all packets starting from the one that caused the error.
3. Advantages: Simplicity of implementation, and efficient use of bandwidth in error-prone networks.
4. Disadvantages: Inefficient use of bandwidth in networks with low error rates, as it requires retransmitting multiple packets even if only one is lost or corrupted.

Selective-Reject ARQ:

1. Overview: Selective-Reject ARQ, also known as Selective Repeat ARQ, is a sliding window protocol that allows the receiver to selectively request retransmission of lost or corrupted packets.
2. Operation: Unlike Go-Back-N ARQ, which requires retransmitting multiple packets upon detecting errors, Selective-Reject ARQ allows the receiver to request retransmission of only the missing or corrupted packets. This minimizes the retransmission overhead and improves bandwidth utilization.
3. Advantages: More efficient use of bandwidth compared to Go-Back-N ARQ, especially in networks with low error rates.
4. Disadvantages: Increased complexity of implementation compared to Go-Back-N ARQ.

Parity Check

1. Overview: Parity check is a simple method used to detect errors in data transmission. It involves adding bit to each data unit so that the total number of 1s in the unit, including the parity bit, is either even or odd.

2. Operation: In even parity, the parity bit is set to ensure that the total number of 1s (including the parity bit) is even. In odd parity, the parity bit is set to ensure that the total number of 1s is odd. Upon reception, the receiver checks if the parity is consistent with the agreed-upon scheme. If not, an error is detected.

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2D Parity:

- Overview: 2D Parity extends the concept of parity check to two dimensions, typically used in memory systems and data storage.

- Operation: Instead of just adding a single parity bit to each data unit, a 2D array of parity bits is calculated. Rows and columns of data units are summed separately, and parity bits are added to ensure that the total number of 1s in each row and column, including the respective parity bits, is either even or odd.

Checksum:

- Overview: Checksum is a method used for error detection in data transmission. It involves summing up the values of data units using a predetermined algorithm and appending the result as a checksum to the data being transmitted.

- Operation: The sender calculates the checksum by summing up the values of data units using an algorithm such as the Internet Checksum algorithm or the Fletcher checksum algorithm. The receiver performs the same calculation on the received data and compares the result with the transmitted checksum. If they differ, an error is detected.

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CRC (Cyclic Redundancy Check):

1. Overview: CRC is an error-detecting code commonly used in digital networks and storage devices to detect accidental changes to raw data.
2. Operation: CRC involves polynomial division, where the data bits are treated as coefficients of a polynomial. The sender appends a CRC to the data, which is computed based on a predetermined polynomial divisor. The receiver performs the same calculation on the received data and checks if the computed CRC matches the received CRC. If not, an error is detected.
3. Advantages: CRC is highly effective in detecting burst errors, which are consecutive bit errors, and it has widespread use in Ethernet, ZIP files, and other data transmission/storage systems.

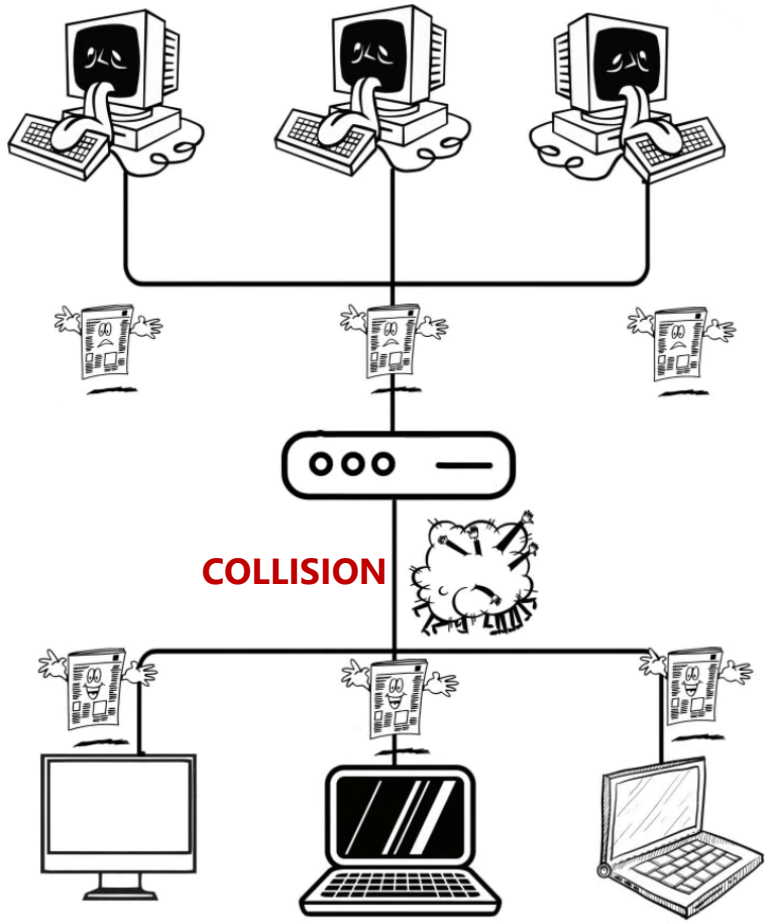
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IEEE 802.1 Data Link Layer

- Two main Sub Layers
- **Logical Link Control (LLC)**
  - Between Network layer & MAC sublayer
- **Medium Access Control (MAC)**
  - Coordinate access to medium
  - Connectionless frame transfer service
  - Machines identified by MAC/physical address
  - Broadcast frames with MAC addresses

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What happens when everybody start sending data ?



Polling:

- Overview: Polling is a method used in computer networking and telecommunications to control the flow of data between devices on a network. It involves a central device, such as a controller or a host, actively querying other devices on the network to determine if they have data to send.

- Operation: In a polling-based system, the central device sequentially polls each connected device, asking if they have data to transmit. If a device has data ready, it sends the data to the central device. Otherwise, the central device moves on to the next device.

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Polling:

Advantages:

1. Simple implementation.
2. Provides centralized control over data transmission.
3. Suitable for networks with a small number of devices and low data traffic.

Disadvantages:

1. Inefficient for networks with a large number of devices or high data traffic, as it introduces polling overhead and may lead to increased latency.
2. Devices may experience idle time waiting for their turn to be polled, reducing overall network efficiency.
3. Vulnerable to issues such as polling collisions and priority inversion

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Token-Passing:

- Overview: Token-passing is a method used in network communication to control access to the communication medium. It involves a token, a special control frame that circulates among the devices on the network. Only the device holding the token is allowed to transmit data.

- Operation: In a token-passing network, the token circulates continuously among the devices. When a device wants to transmit data, it waits for the token to arrive, seizes the token, and then sends its data. After transmitting its data, the device releases the token, allowing it to continue circulating to the next device.

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Token-Passing:

Advantages:

- Provides fair access to the communication medium, as each device gets a turn to transmit data when it holds the token.
- Efficient for networks with a large number of devices or high data traffic, as it eliminates polling overhead and minimizes collisions.
- Guarantees a bounded transmission delay, as the token circulates continuously, ensuring that each device gets a turn within a predictable timeframe.

Disadvantages:

- Complexity of implementation compared to polling-based methods.
- Requires synchronization among devices to ensure proper token handling, which can be challenging in dynamic or heterogeneous networks.
- Susceptible to issues such as token loss or token starvation if a device fails or becomes inactive.

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CSMA/CD (Carrier Sense Multiple Access with Collision Detection):

1. Overview: CSMA/CD is a protocol used in Ethernet networks to control access to the transmission medium and handle collisions. It is primarily employed in wired Ethernet networks.
2. Operation:
    - Carrier Sense: Before transmitting data, a device listens to the network to detect whether the transmission medium is busy. If the medium is idle, the device can proceed to transmit its data.
    - Multiple Access: Multiple devices on the network contend for access to the medium. If two or more devices attempt to transmit simultaneously, a collision occurs.
    - Collision Detection: While transmitting data, a device continuously monitors the network for collision signals. If a collision is detected (i.e., if the transmitted signal doesn't match the received signal), the device stops transmitting immediately and initiates a backoff algorithm to retransmit the data after a random time interval.

CSMA/CD (Carrier Sense Multiple Access with Collision Detection):

1. Advantages:
   - Efficient utilization of the network medium.
   - Fair access to the network for all devices.
   - Ability to detect and handle collisions, minimizing data loss and retransmissions.

2. Disadvantages:
   - Limited scalability in large networks with high collision rates.
   - Increased overhead due to collision detection and retransmission mechanisms.
   - Inefficiency in networks with long propagation delays, as it takes time to detect collisions.

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CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance):

1. Overview: CSMA/CA is a protocol used in wireless networks to control access to the shared wireless medium and avoid collisions. It is commonly used in Wi-Fi networks.
2. Operation:
    - Carrier Sense: Similar to CSMA/CD, devices listen to the wireless medium to detect ongoing transmissions. If the medium is busy, the device waits for it to become idle before attempting to transmit.
    - Multiple Access: Multiple devices contend for access to the wireless medium. To avoid collisions, devices employ a "listen before talk" approach, where they defer transmission if they detect other devices transmitting.
    - Collision Avoidance: Devices use techniques such as the Request to Send (RTS) and Clear to Send (CTS) handshaking mechanism to reserve the medium before transmitting data. This helps in reducing the likelihood of collisions by allowing devices to reserve the medium for a specified duration.

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance):

Advantages:

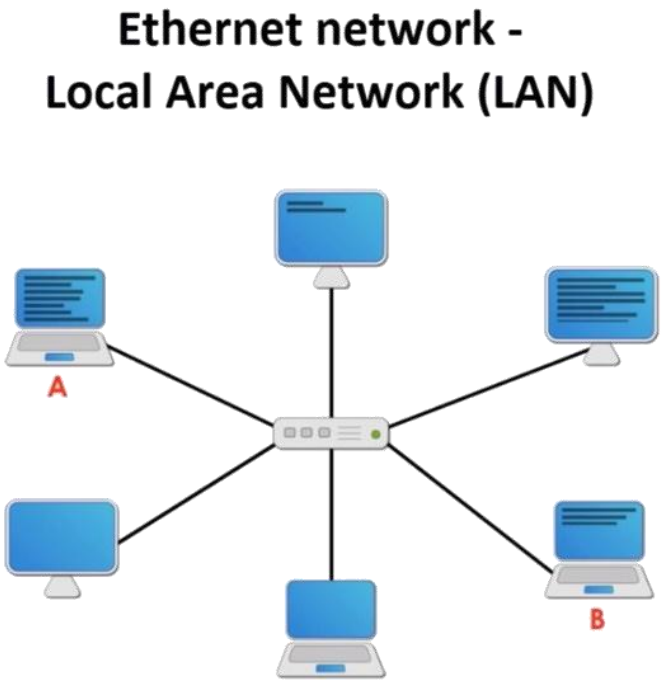
- Effective in reducing collisions in wireless networks.
- Supports a higher degree of reliability and predictability compared to CSMA/CD in wireless environments.
- Suitable for environments with variable channel conditions and interference.

Disadvantages:

- Increased overhead due to the RTS/CTS handshaking mechanism.
- Susceptibility to hidden node and exposed node problems, which can affect network performance.
- Reduced throughput compared to wired networks due to overhead and contention.

What is LAN?

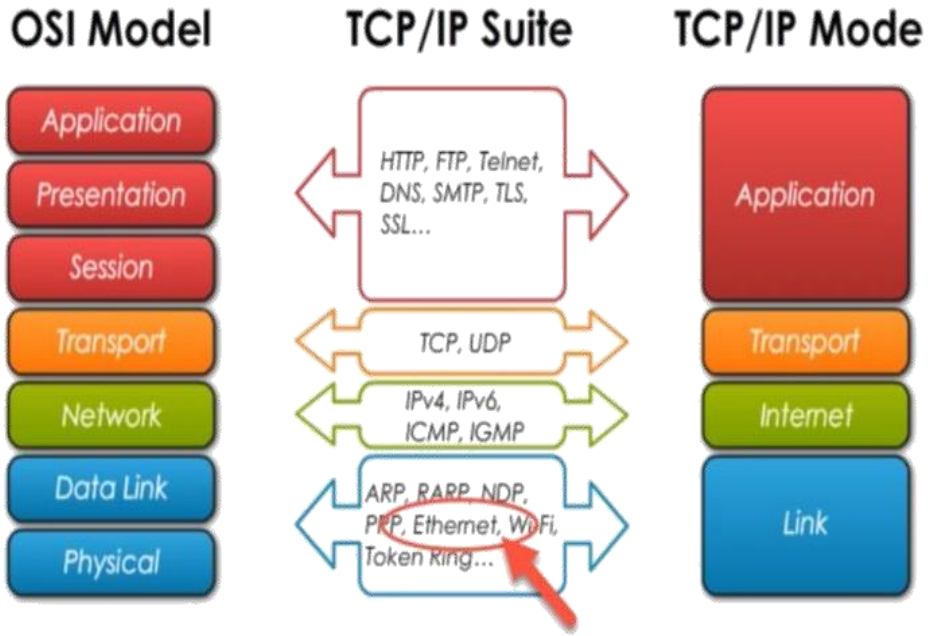
- **Private ownership**
  - freedom from regulatory constraints of WANs
- **Short distance** (~1km)
  - low cost
  - very high-speed, relatively error-free communication
  - complex error control unnecessary
- **LAN** characterizes:
  - Topology (Star, Bus, Mesh, etc.)
  - Protocols (CSMA/CD, CSMA/CA, etc.)
  - Media (twisted pair, coaxial, fiber optic)



Ethernet

- **LAN Technology** (most popular)
- Other LAN technologies
  - Token Ring
  - FDDI (fiber distributed data interface)
  - ARCNET

- Operates at both **physical** and **data link layer**



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Collision vs. Broadcast Domain

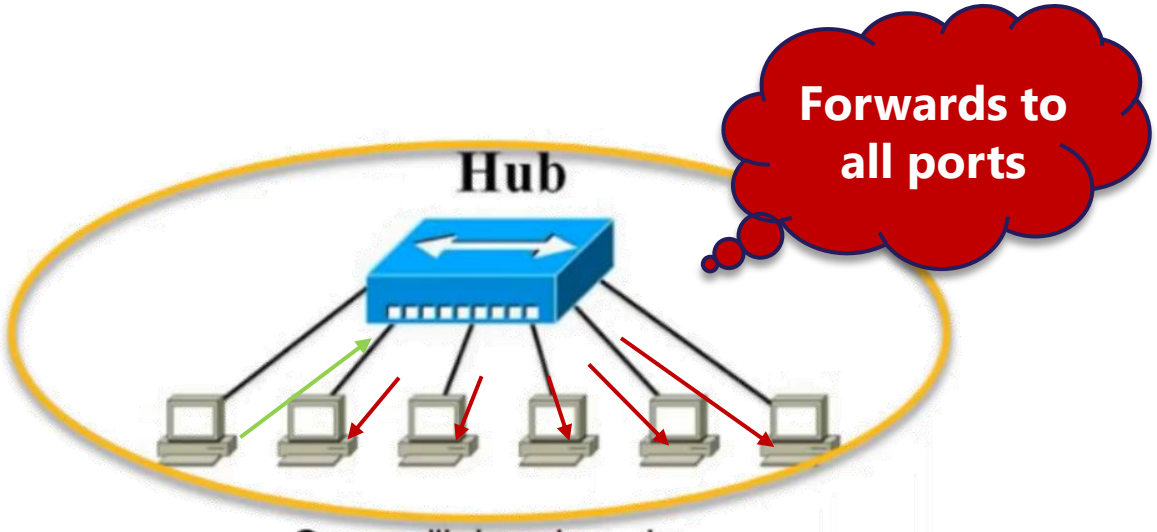
- **Collision Domain**
  - A group of nodes that can hear each other
  - A part of a network where packet collisions can occur
- **Broadcast Domain**
  - Contains devices that can reach each other at the data link layer by using **broadcast**



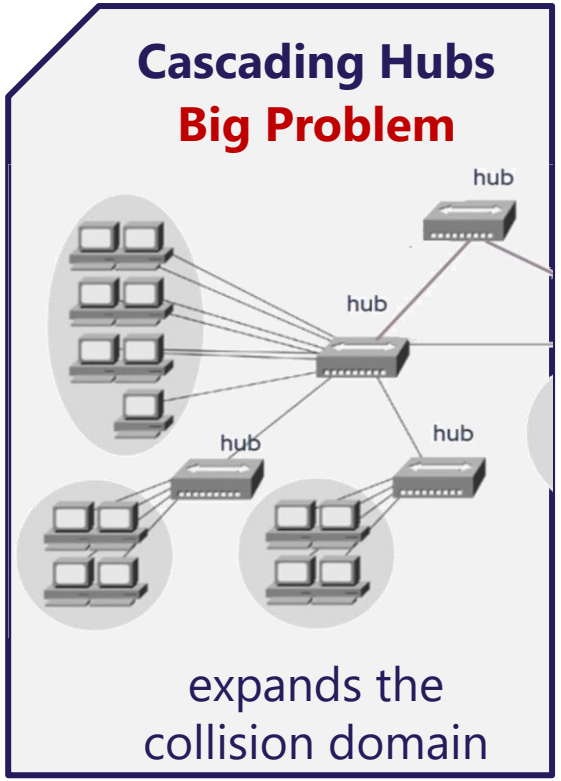
- **A broadcast stops at the router**
  - It removes the header of the frame including the destination MAC address with another destination MAC address to send it to another broadcast domain

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Layer 01: Hub



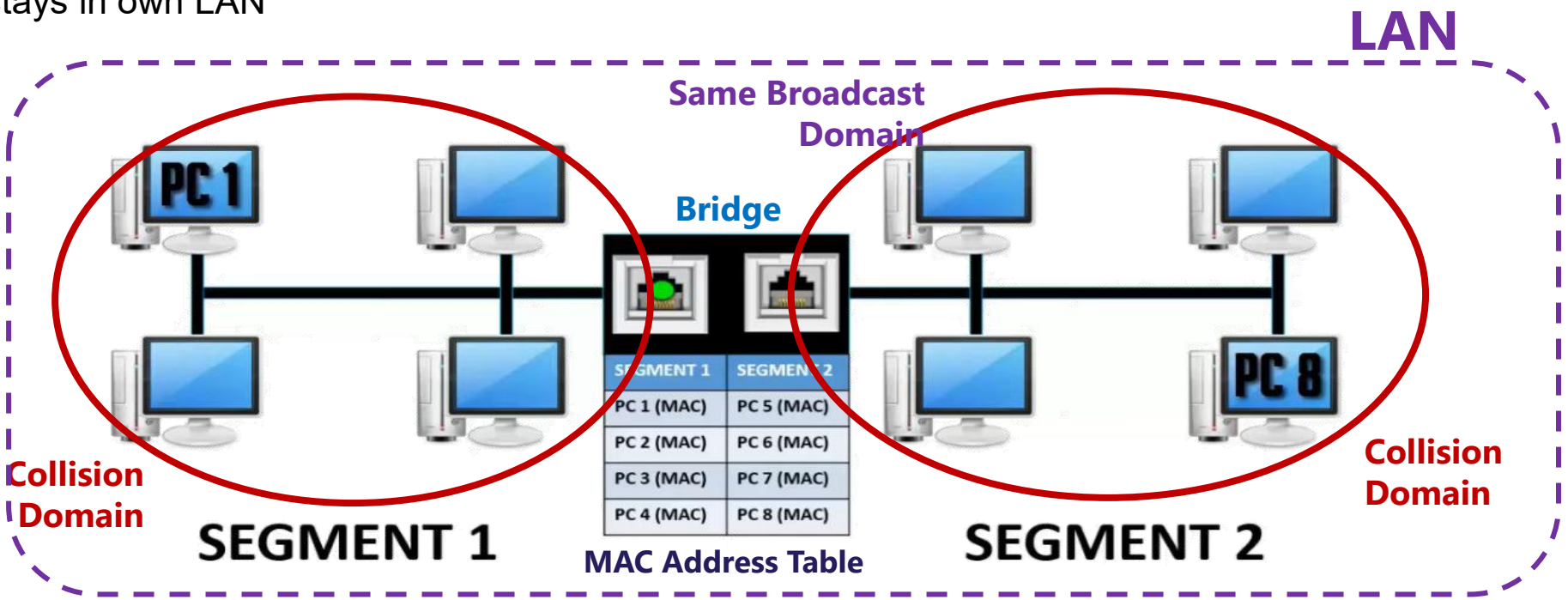
Only operate in **Half Duplex** mode



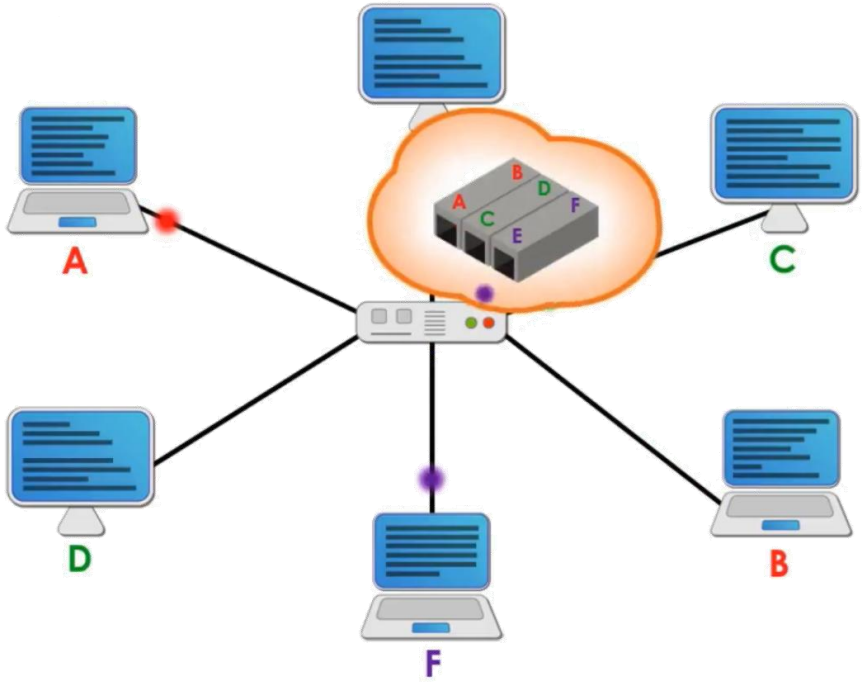
Layer 02: Bridge

- Fewer Ports
- Replaced by Switch

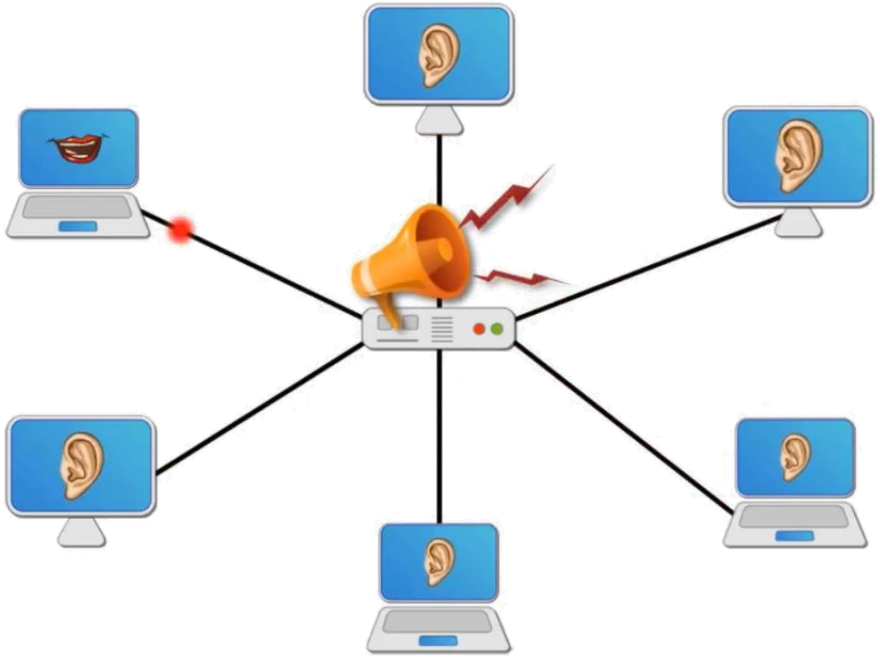
- **Segments a LAN**
  - MAC address filtering
  - Local traffic stays in own LAN



Layer 02: Switch



Each port is a collision domain

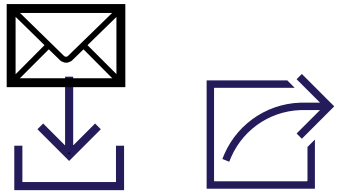


One broadcast domain

Can operate in **Full Duplex** Mode

Layer 02: Switch – cont.

- **Switch buffers** network **frames**. **(store-and-forward)**
  - **Forward later** when the egress link is idle



This decoupling of receive and transmit operations enables a network that are largely independent from each other and only compete for link bandwidth.

  - **On a collision:**
    - Any collision will not propagate across the switch
    - Switch will retransmit the buffered frame later-on

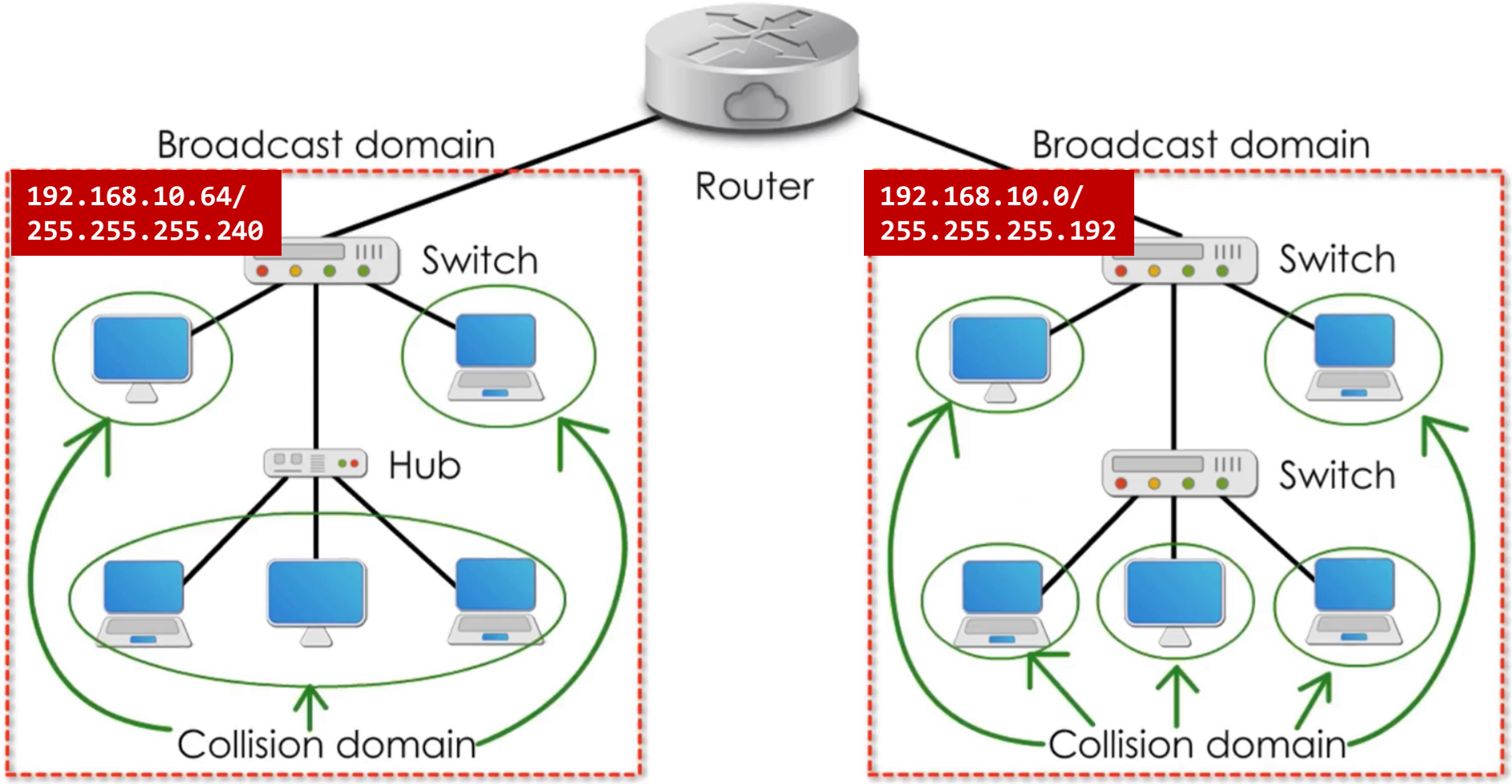
Switch segregates collision domains (on half-duplex links) or removes them completely on full-duplex links

Layer 02: Switch – cont.

In contrast:

- **Hubs** repeat incoming bits as they are received **(No Buffering)**
- **On a collision:**
  - A collision on an egress interface will disrupt reception on the ingress interface
  - The hub needs to propagate an upstream collision back to the source
  - All nodes connected to a hub (or potentially chained hubs) form a single, common collision domain
- **Only one** node at a time **can transmit**

Layer 03: Router



Thank you

